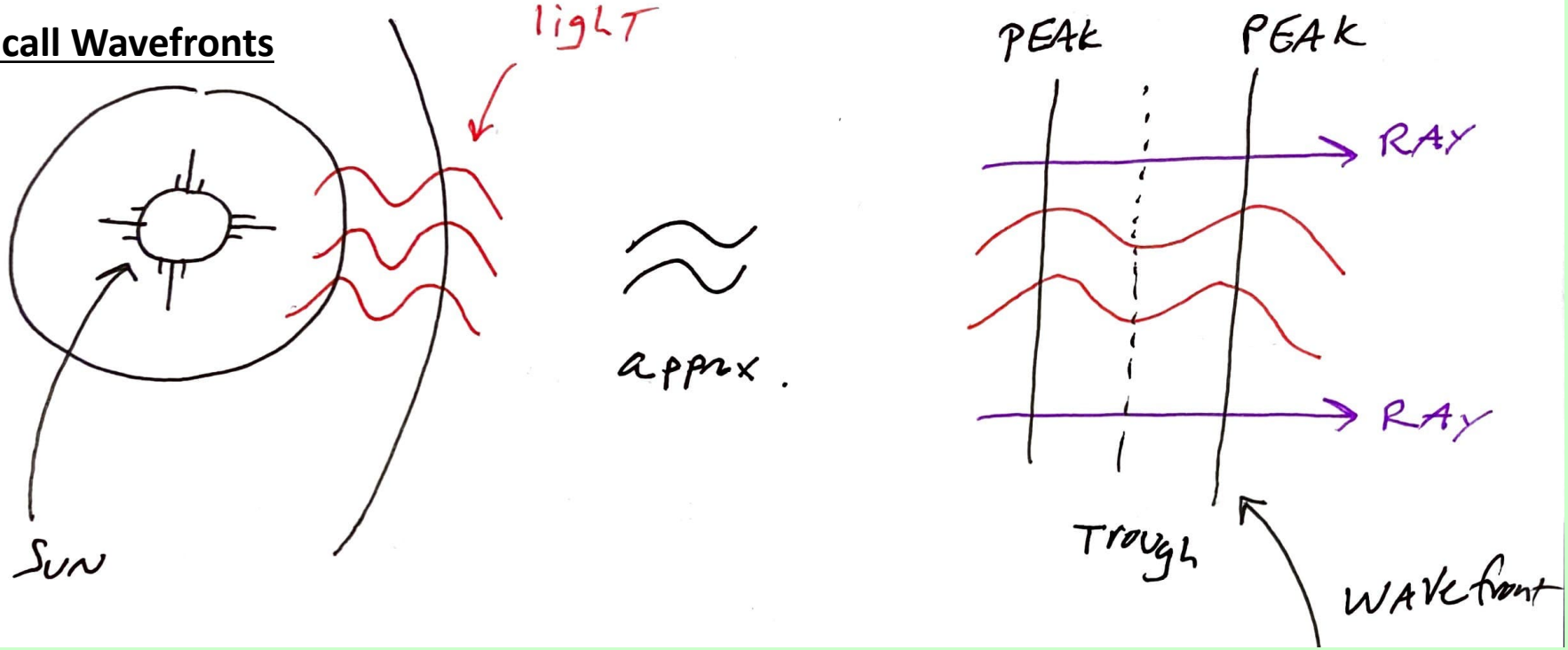
Reflection and Refraction

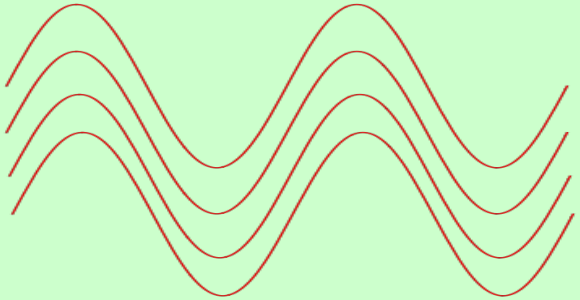


Recall Wavefronts



Ex. Convert to a wavefront model

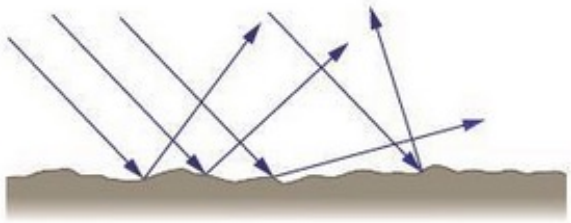
Label: Peaks, Troughs, Rays



Reflection= “Bouncies”

Why are mirrors shiny?

Diffuse Reflection

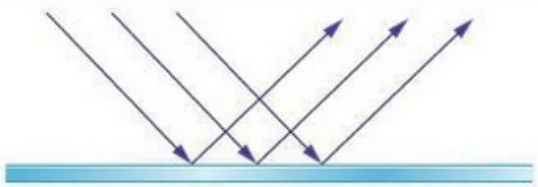


(a) Diffuse reflection



(b) No image with diffuse reflection

Specular Reflection



(a) Specular reflection

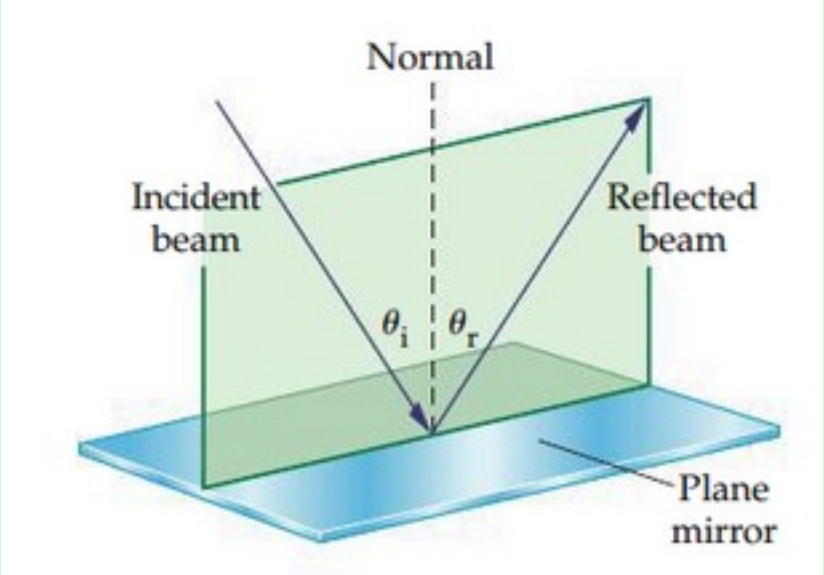


(b) Mirror image due to specular reflection

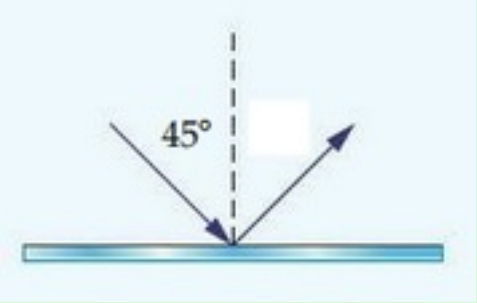
Ex. What in this room is specular/diffuse?

Law of Reflection:

$$\theta_i = \theta_r$$



Ex. What is the reflected angle?

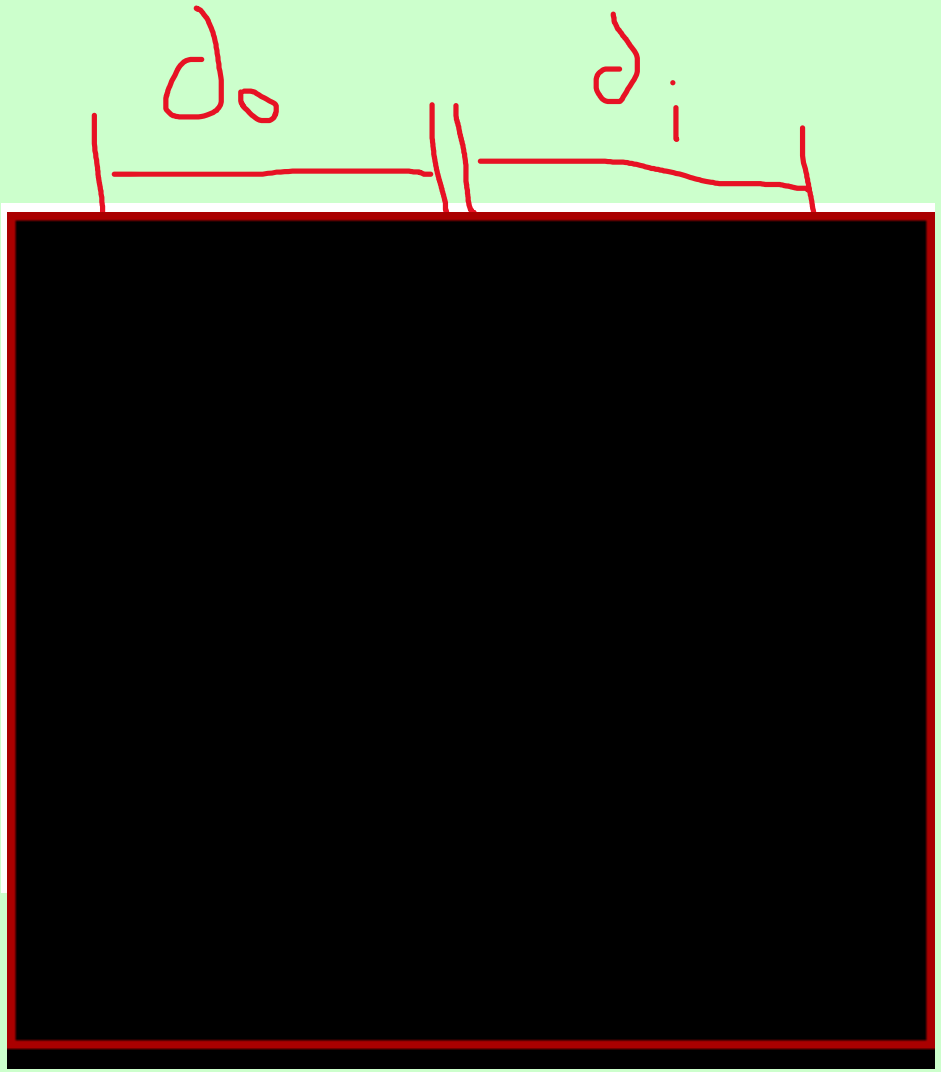


Demo: Laser with mirror

Mirror: Brain assumes straight lines

Where rays converge, an **image** is formed

Real: places rays go
Virtual: Places they appear to go



d_0 Is the object distance

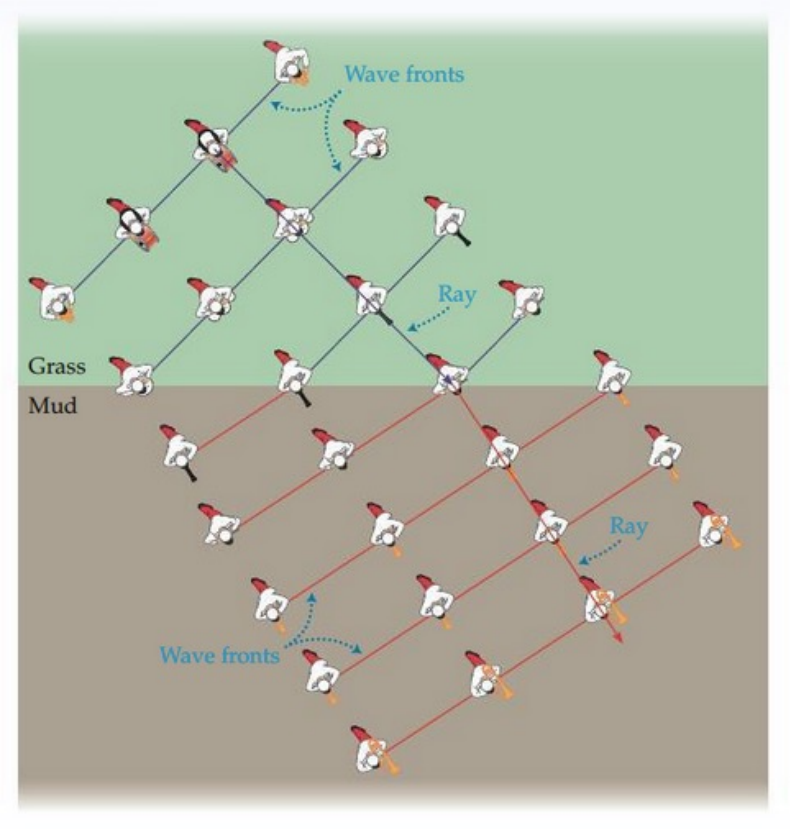
d_i Is the image distance

(for a flat mirror) $$|d_0| = |d_i|$$

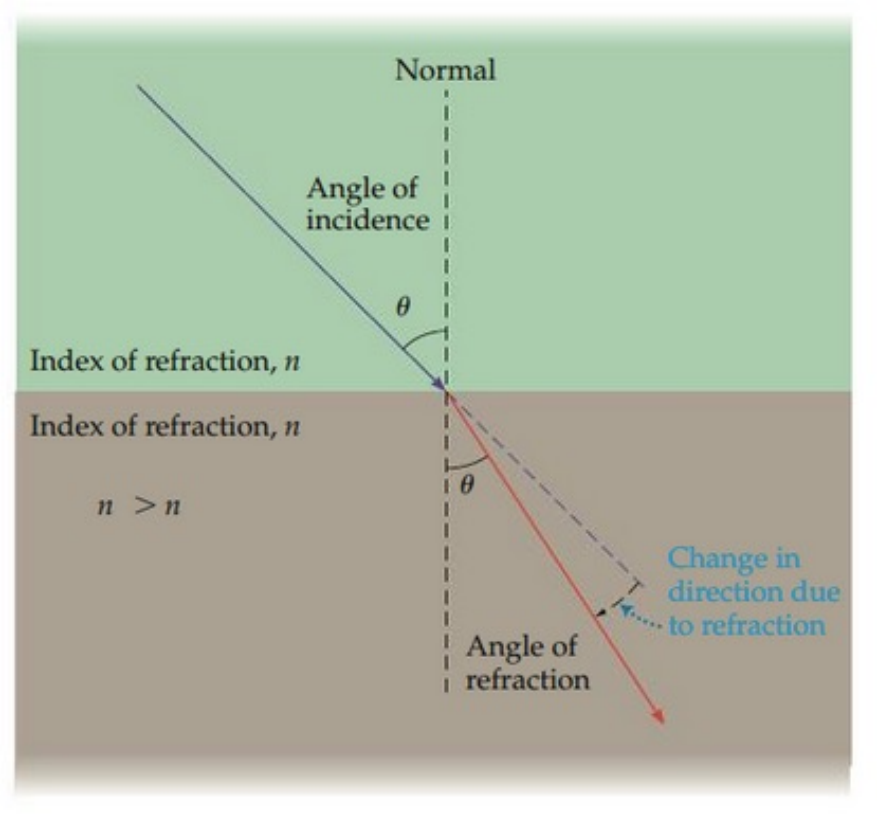
Refraction="Bendies"

Snell's law:

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$



Marching band lines bend when entering slower ground



Light bends when entering a new medium

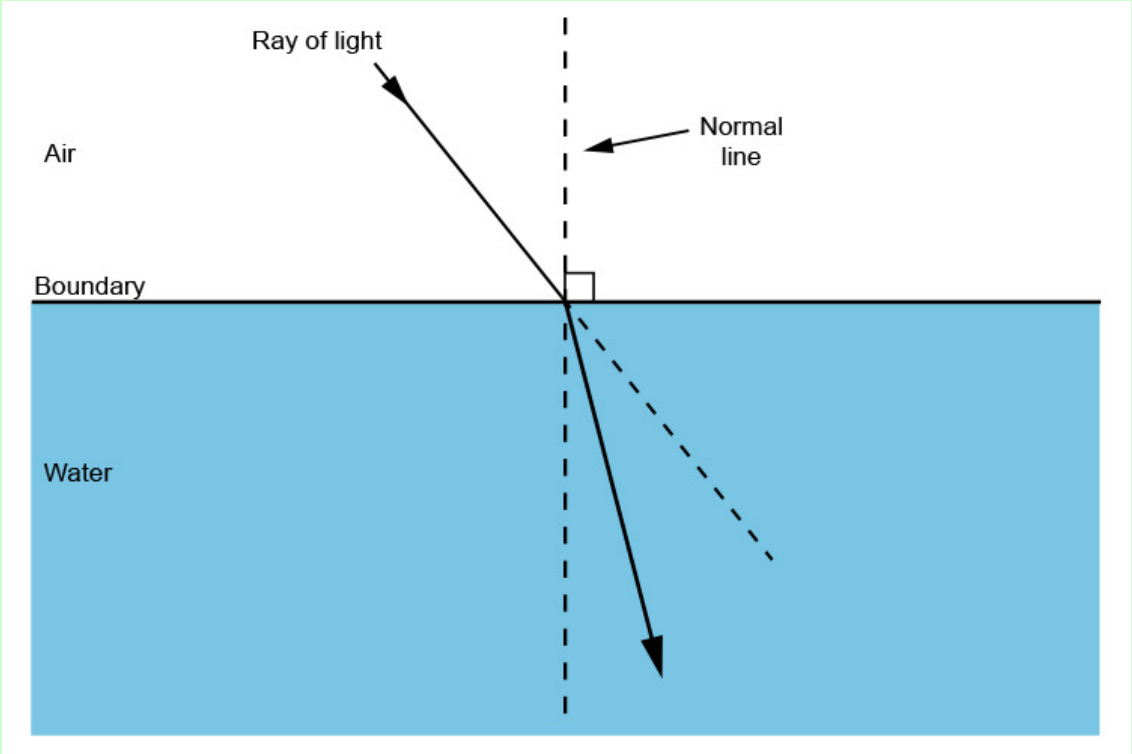
If n1<n2: Bend is **toward** normal
If n1>n2: Bend is **away from** normal

Ex. Find the refracted angle

$$\theta_1 = 30^\circ$$

$$n_{\text{water}} = 1.33$$

$$n_{\text{air}} = 1.00$$



(optional derivation of Snell's Law)

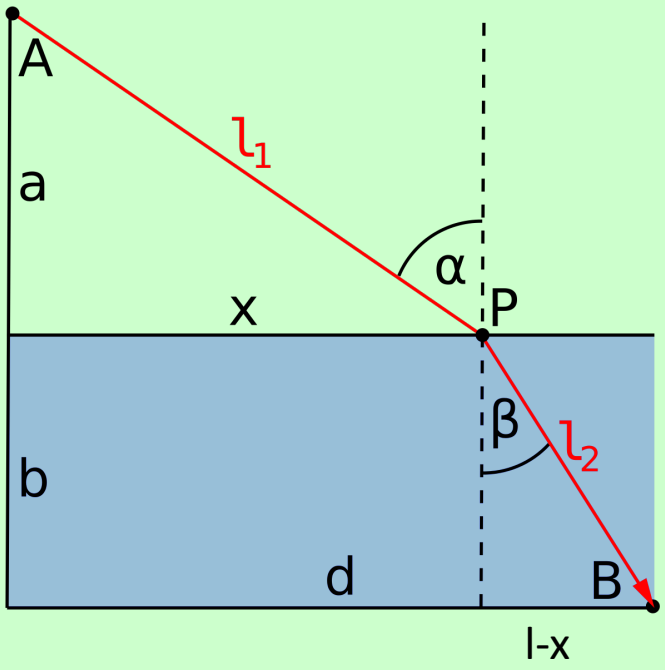
Via Fermat's Principle

$$l_1 = \sqrt{x^2 + a^2} \qquad n = \frac{c}{v}$$

$$l_2 = \sqrt{(l-x)^2 + b^2}$$

$$t = \frac{d}{v} = \frac{l_1}{v_1} + \frac{l_2}{v_2} \qquad t = \frac{n_1 l_1}{c} + \frac{n_2 l_2}{c}$$

$$t = \frac{n_1\sqrt{x^2 + a^2}}{c} + \frac{n_2\sqrt{(l-x)^2 + b^2}}{c}$$

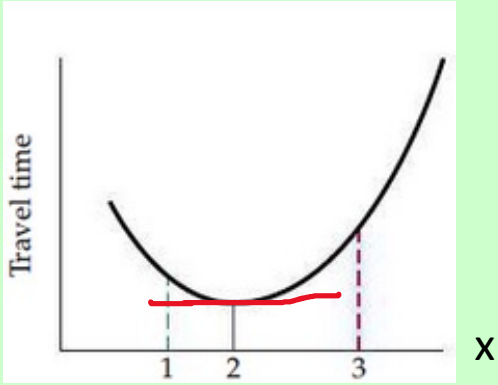


At shortest travel time, derivative is zero

$$\frac{dt}{dx} = 0 = \frac{d}{dx}\left(\frac{n_1\sqrt{x^2 + a^2}}{c} + \frac{n_2\sqrt{(l-x)^2 + b^2}}{c}\right)$$

$$0 = \frac{n_1 x}{c\sqrt{x^2 + a^2}} - \frac{n_2(l-x)}{c\sqrt{(l-x)^2 + b^2}}$$

Snells law $\boxed{0 = n_1 \sin\alpha - n_2 \sin\beta}$



Bottom of function when d/dx=0

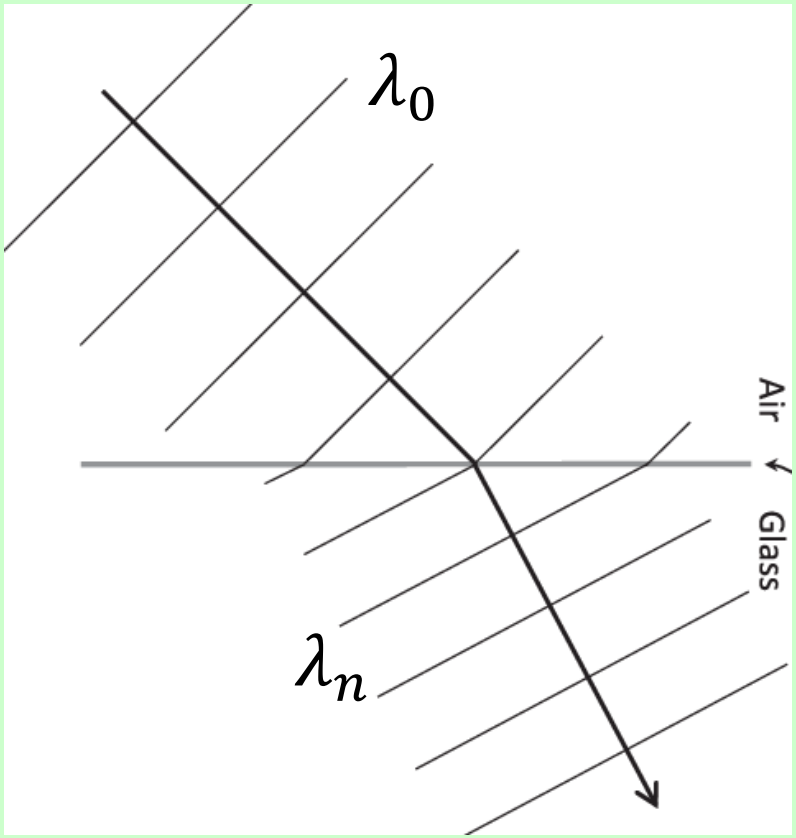
Wavefront Analysis

$$v = \lambda_n f$$

$$n = \frac{c}{v} \implies \boxed{v = \frac{c}{n}}$$

$$\frac{c}{n} = \lambda_n f$$

$$\lambda_n = \frac{c}{nf}$$



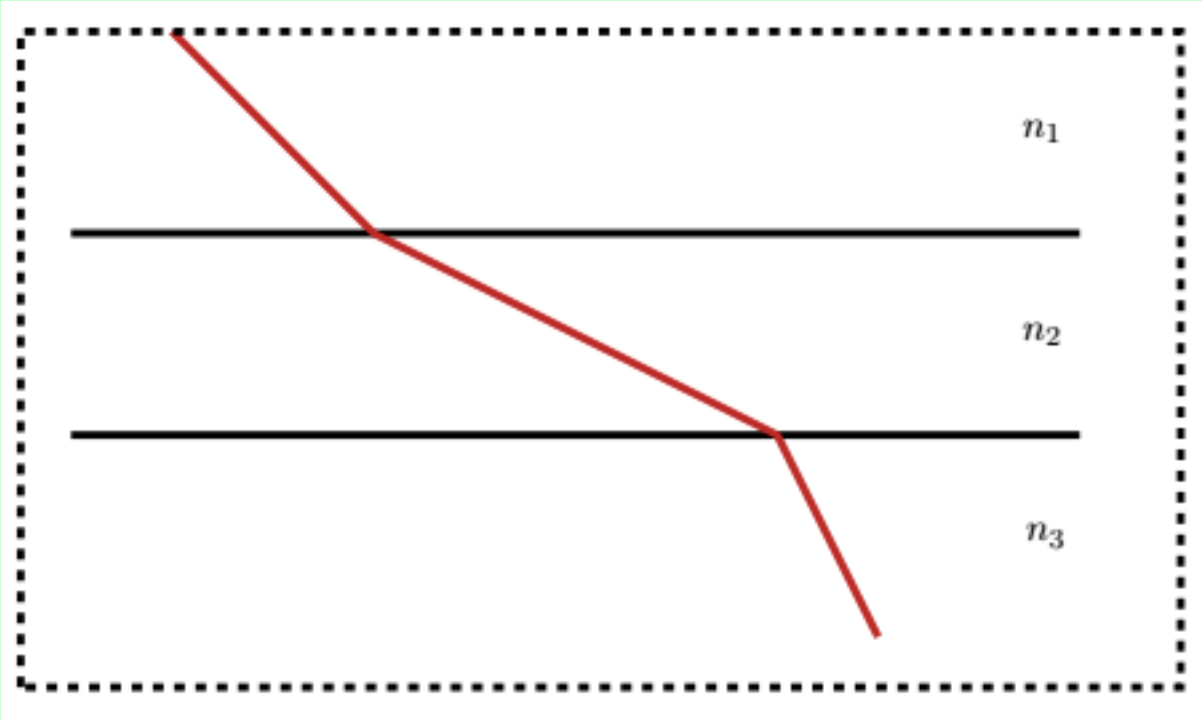
C and f are both constant!!

$$\lambda_n \propto \frac{1}{n}$$ Or $$\boxed{\lambda_n = \frac{\lambda_0}{n}}$$

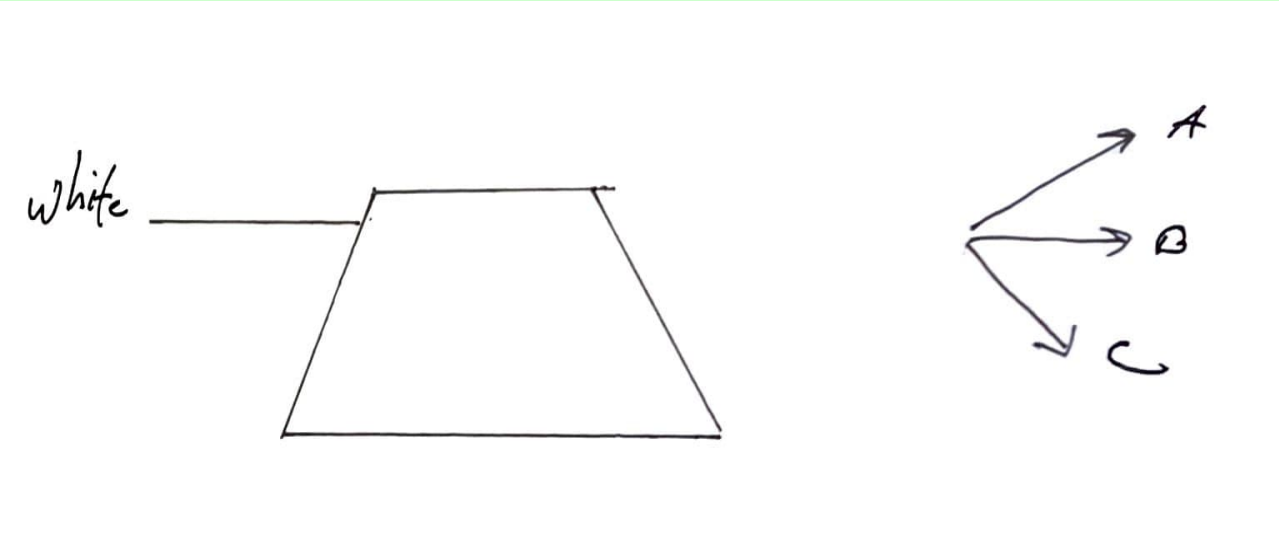
Wavelength dependends on n

Ex. 400 nm light travels from air to glass (n = 1.5).
What are: λ_n , f_n , v_n in the glass?

Ex. Rank according to n

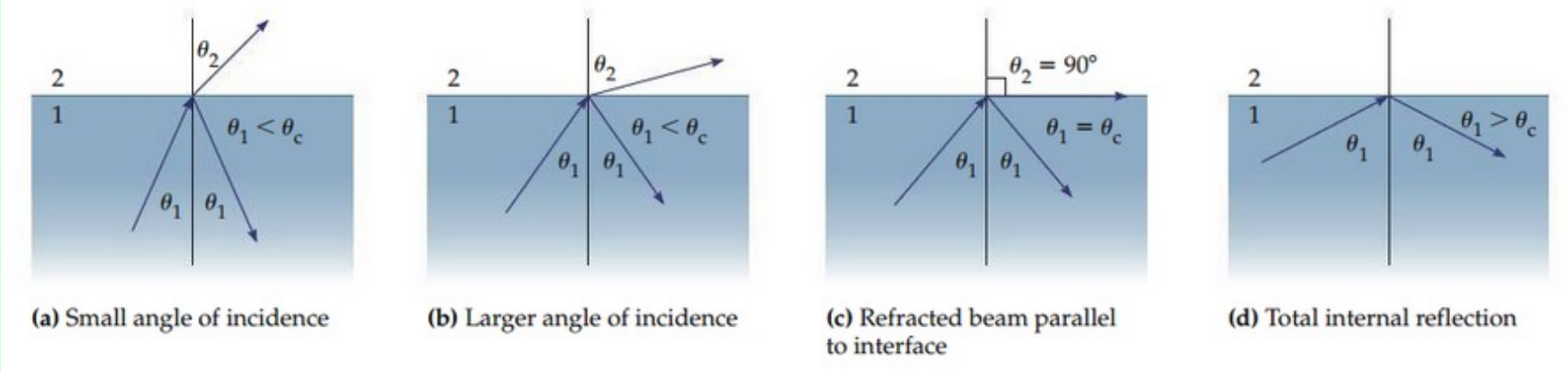


Ex. In which direction will the beam emerge?



Total Internal Reflection

θ_r can't be $> 90°$



Critical angle

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

$$n_1 \sin\theta_c = n_2 \sin(90)$$

$$\boxed{\theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right)}$$

Ex. What is the critical angle for a scuba diver shining his flashlight up toward the surface?

Demos: optical fiber, H20 Can, Prism

Dispersion

$v = \lambda f$

$n_\lambda = \frac{c}{v} = \frac{c}{\lambda f}$ ← const.

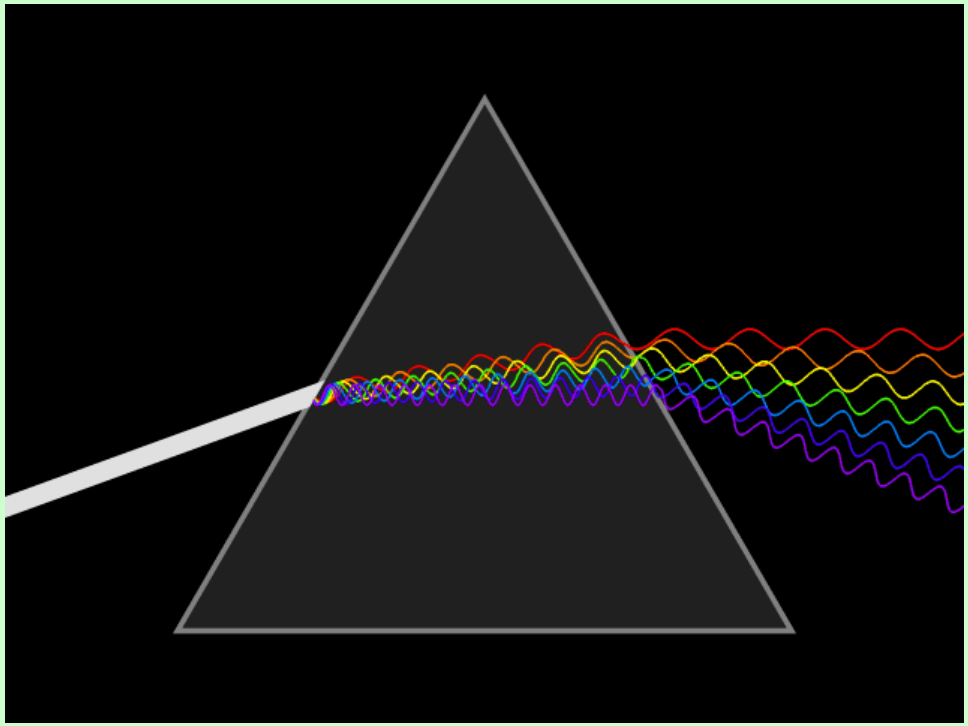
$n_\lambda \propto \frac{1}{\lambda}$ blue bends more!!

Crown glass

$n_\lambda = 1.54$ (blue)

$n_\lambda = 1.51$ (red)

Ex. Trace light through a prism and show that this happens



Why Rainbows?

